

Task 10: Clustering using K-Means

Introduction: K-Means is an unsupervised machine learning algorithm used to group similar data points into clusters. It is widely used for pattern recognition and data segmentation.

Working of K-Means:

1. Choose the number of clusters (K).
2. Initialize K centroids randomly.
3. Assign each data point to the nearest centroid.
4. Recalculate centroids based on assigned points.
5. Repeat until convergence (no change in clusters).

Centroids: Centroids are the central points of clusters. They represent the average position of all data points within a cluster.

Choosing K (Elbow Method): The Elbow Method helps determine the optimal number of clusters by plotting the Within-Cluster Sum of Squares (WCSS) against K. The point where the graph bends (elbow) is the optimal K.

Dataset: Mall Customers dataset is commonly used. It contains features like Annual Income and Spending Score.

Implementation Steps:

- Load dataset using pandas
- Select relevant features
- Apply KMeans from scikit-learn
- Use Elbow method to find optimal K
- Fit model and predict clusters
- Visualize clusters and centroids

Evaluation: Since K-Means is unsupervised, evaluation is done using visualization and metrics like inertia (WCSS).

Conclusion: K-Means is simple and efficient for clustering tasks but requires choosing the right K and is sensitive to initial centroid placement.